

GPU Batch Lunar Gravity Validation

Fixed-step GPU propagation vs an adaptive reference

2026-05-29 13:22 UTC

Parameter	Value
Scenarios	100
Sampling method	random
Scenario seed	42
Scenario mode	near_circular_altitude
Inclination sampling	uniform_deg
Altitude range	100 - 1000 km
Duration	5 days
GPU integrator	medium (fixed step 30 s)
Truth workers	4
Output cadence	60 s
Precision	float32
Truth	SH200 DOP853
GPU models	sh20,sh60,st_lrps
Frame mode	match_dynamics_engine

Reference note: the truth trajectories are a high-accuracy adaptive numerical reference, not physical ground truth. GPU fixed-step models carry both model and integration error.

Executive Summary

Metric	Value
Most accurate GPU model	GPU_ST_LRPS_RK4
Fastest GPU model	GPU_SH20_RK4
ST-LRPS closest by median RMS	GPU_SH60_RK4
ST-LRPS median-equivalent status	better_than_sh60
ST-LRPS closest by P95 RMS	GPU_SH60_RK4
Best ST-LRPS scenario	369
Representative ST-LRPS scenario	437
Worst ST-LRPS scenario	366

Accuracy Ranking

Sorted best-to-worst by median RMS position error.

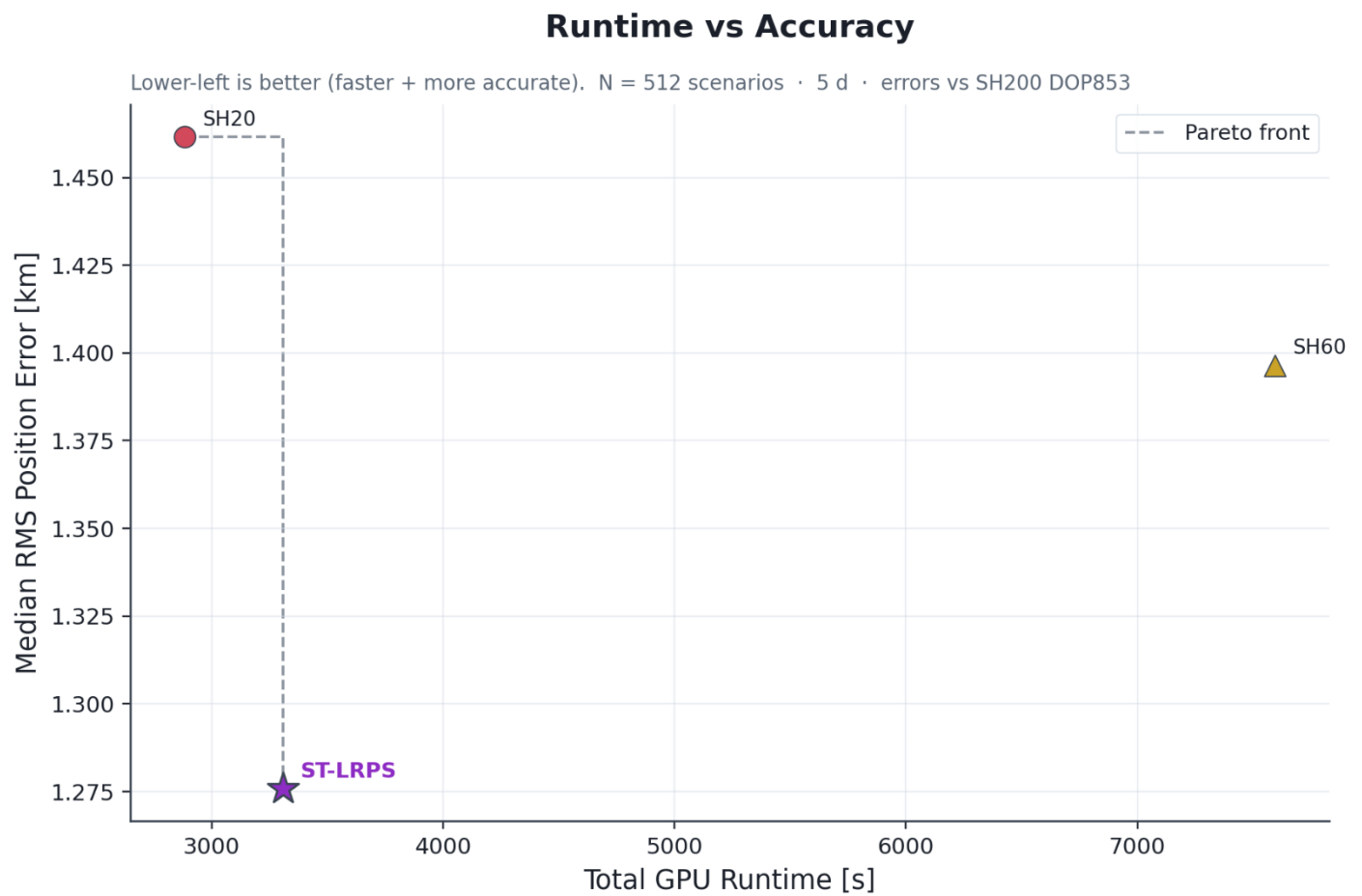
Model	Median RMS [km]	P95 RMS [km]	Max RMS [km]	Median Along [km]
GPU_ST_LRPS_RK4	1.2759	3.7266	5.9879	1.2742
GPU_SH60_RK4	1.3965	3.9270	5.9512	1.3952
GPU_SH20_RK4	1.4615	4.1992	6.1544	1.4598

Runtime

Wall-clock runtime and throughput for the GPU fixed-step propagation.

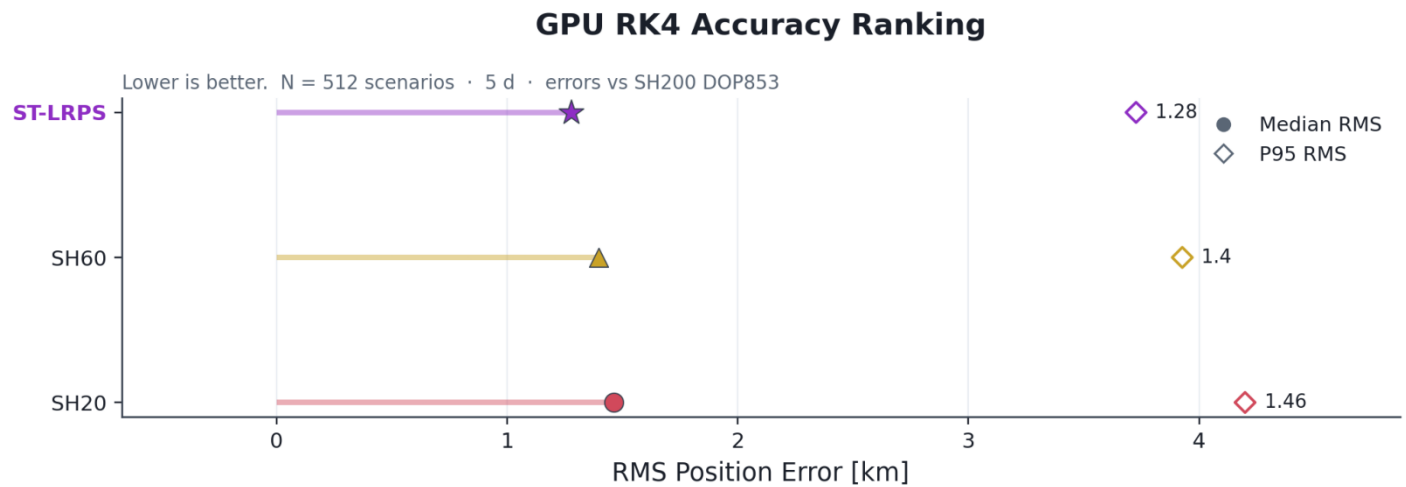
Model	Total [s]	Per scenario [s]	Steps/s	Speedup vs truth
GPU_SH20_RK4	2885.103	5.63497	1277.7	39.71
GPU_ST_LRPS_RK4	3308.686	6.46228	1114.2	34.63
GPU_SH60_RK4	7593.295	14.83065	485.5	15.09

Figure



Runtime-accuracy tradeoff across GPU models. Lower-left is better (faster and more accurate); the dashed line marks the Pareto front. N = 100 scenarios over 5 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

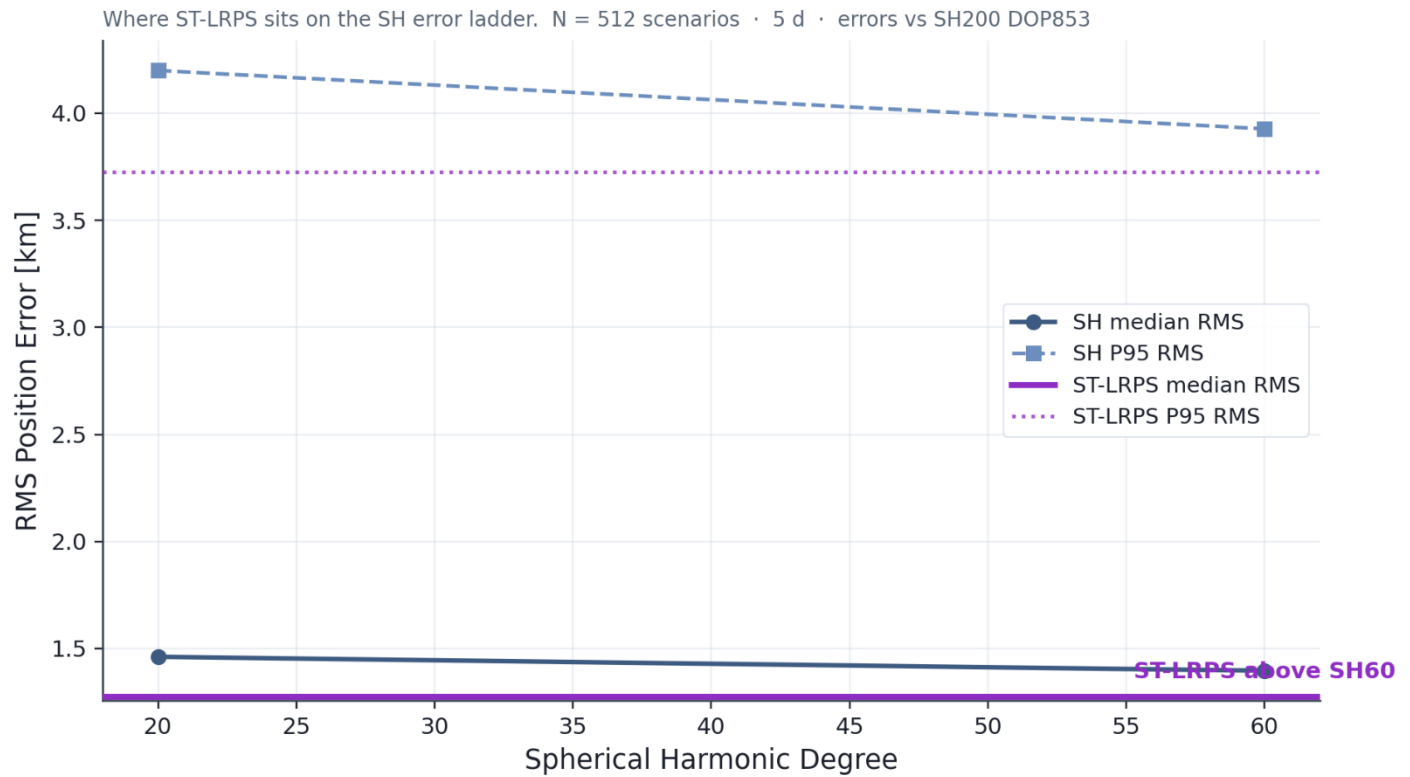
Figure



Per-model accuracy ranking (lollipops = median RMS, open diamonds = P95 RMS), sorted best-to-worst; ST-LRPS is highlighted. Lower is better. N = 100 scenarios over 5 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

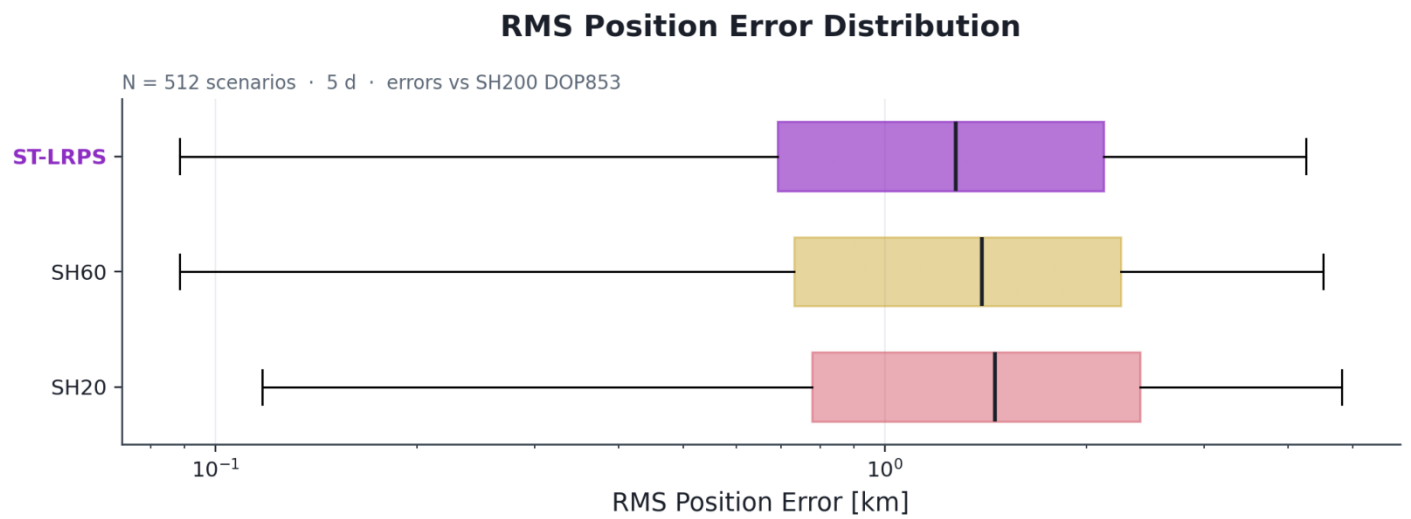
Figure

ST-LRPS Equivalent Spherical-Harmonic Degree



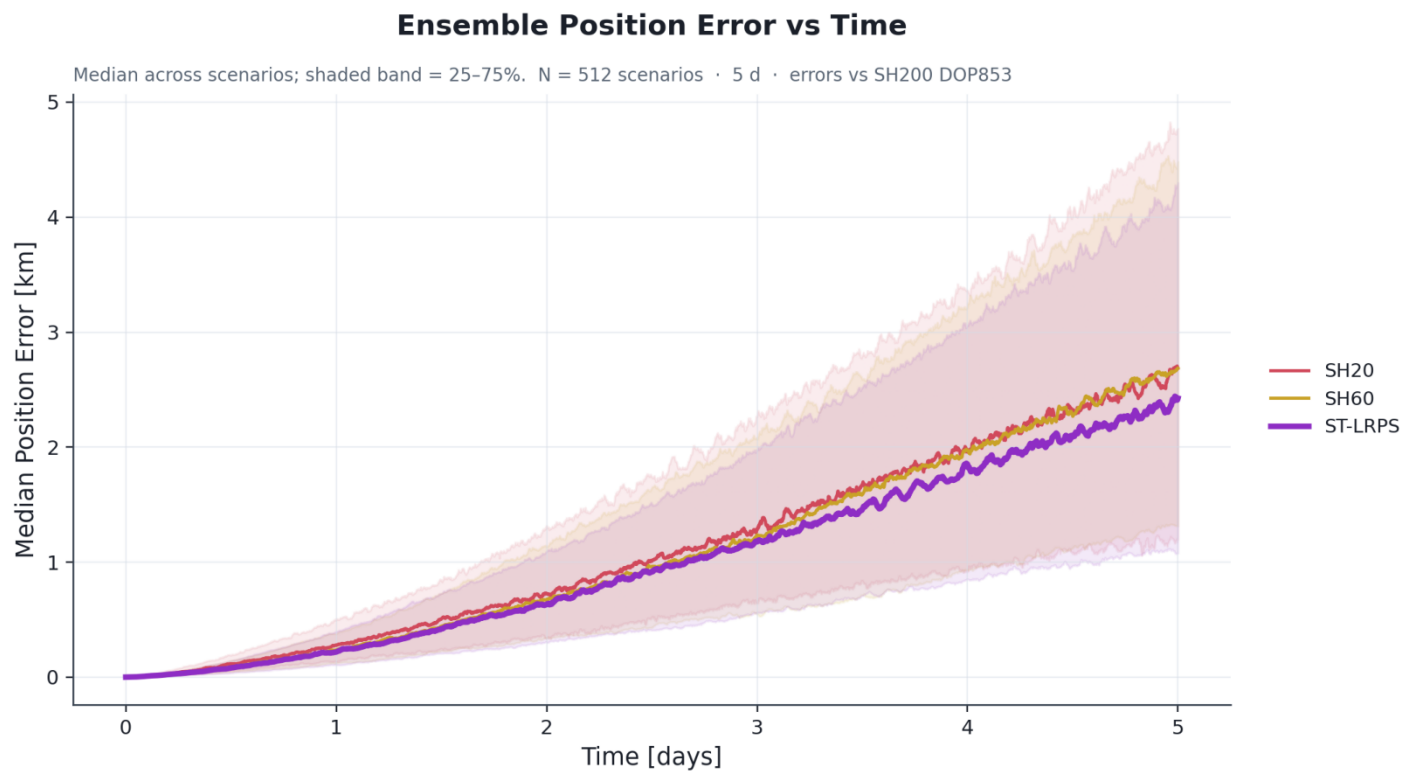
ST-LRPS equivalent SH-degree estimate by interpolating median RMS error across the spherical-harmonic baselines. N = 100 scenarios over 5 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



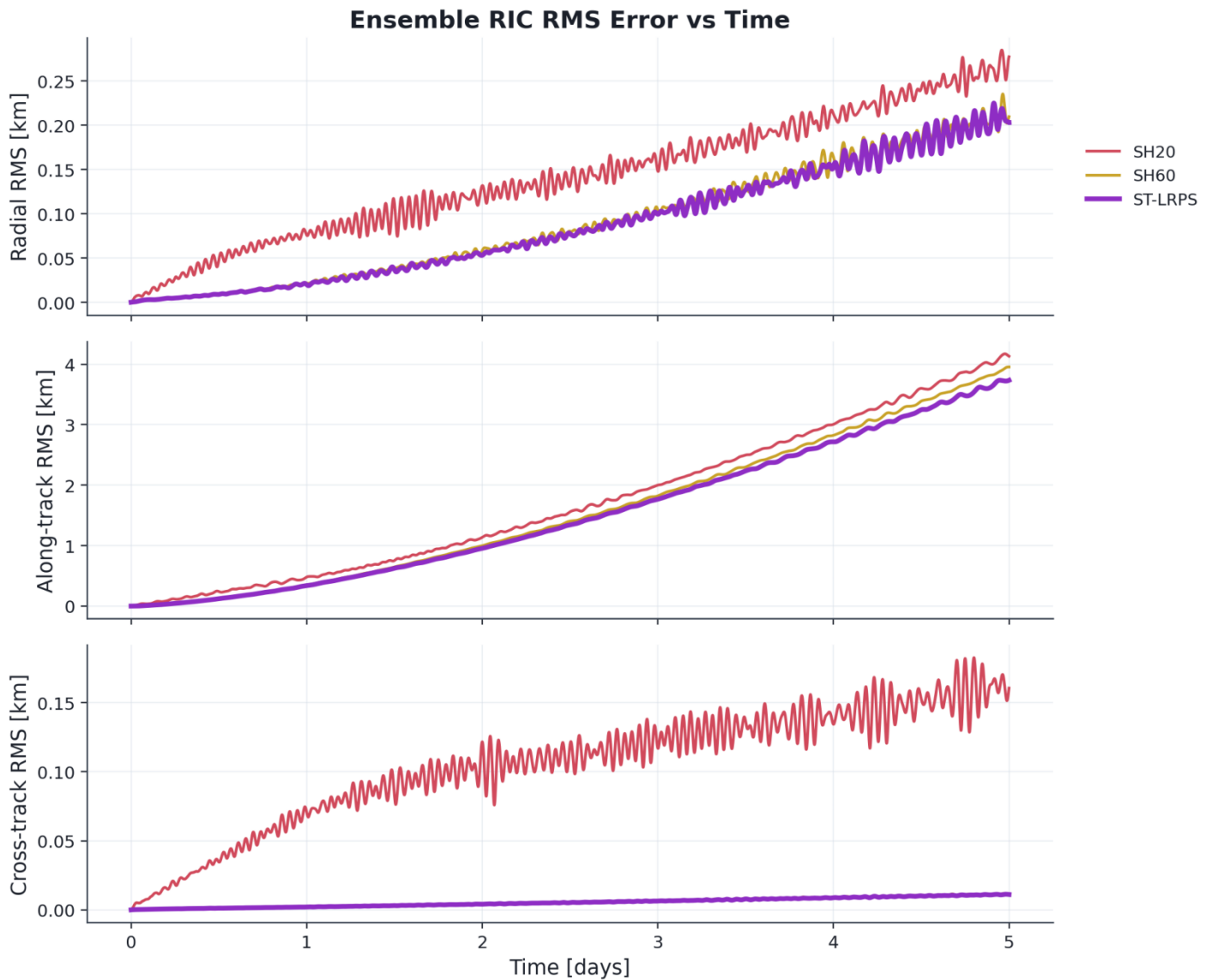
Distribution of per-scenario RMS position error per model. N = 100 scenarios over 5 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



Ensemble median position error vs time (shaded band = 25-75% across scenarios). N = 100 scenarios over 5 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

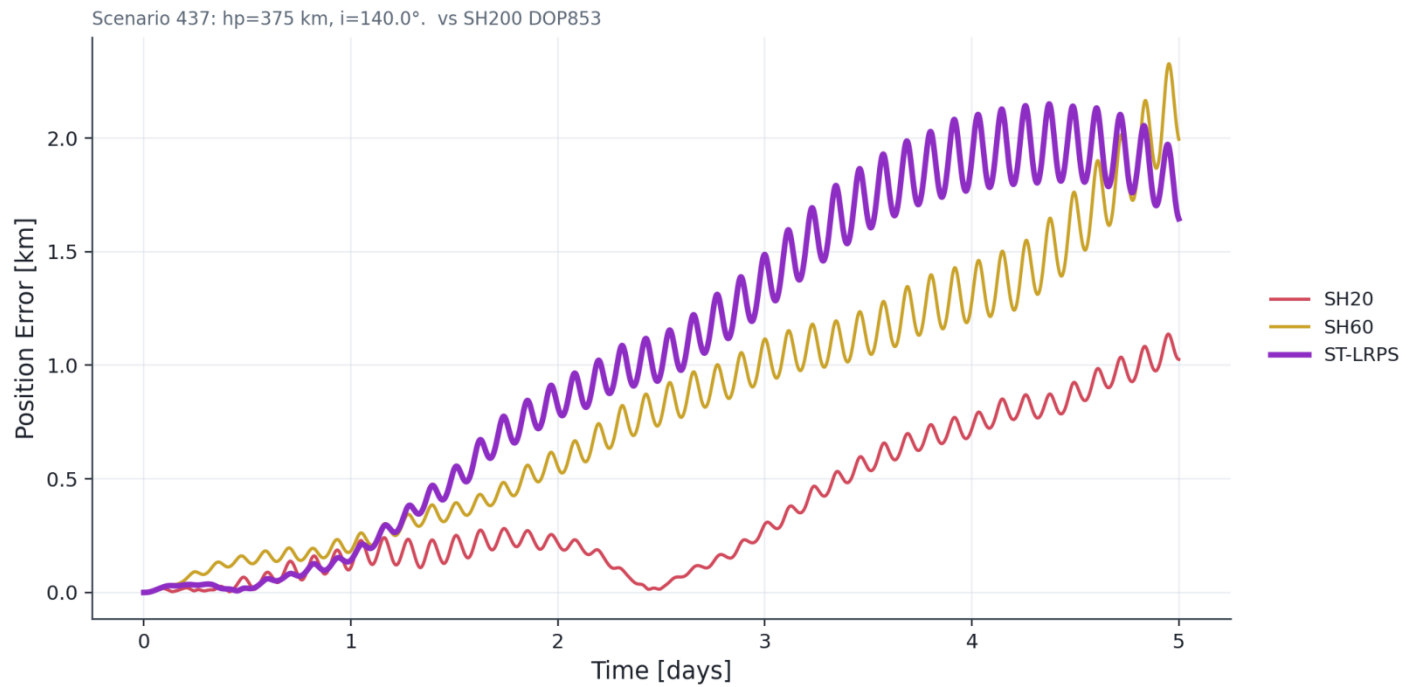
Figure



Ensemble radial/along-track/cross-track RMS error vs time. $N = 100$ scenarios over 5 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

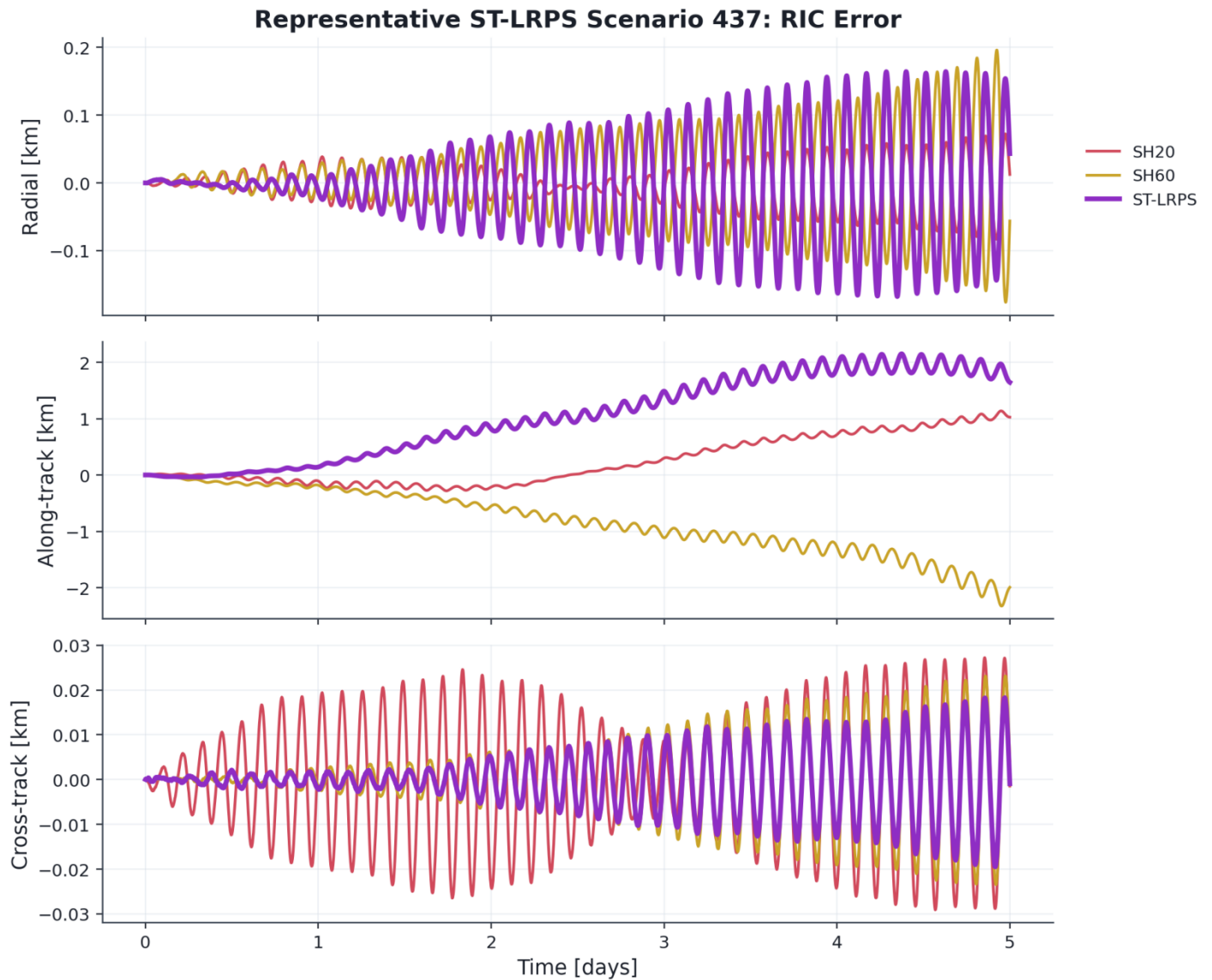
Figure

Representative ST-LRPS Scenario: Position Error



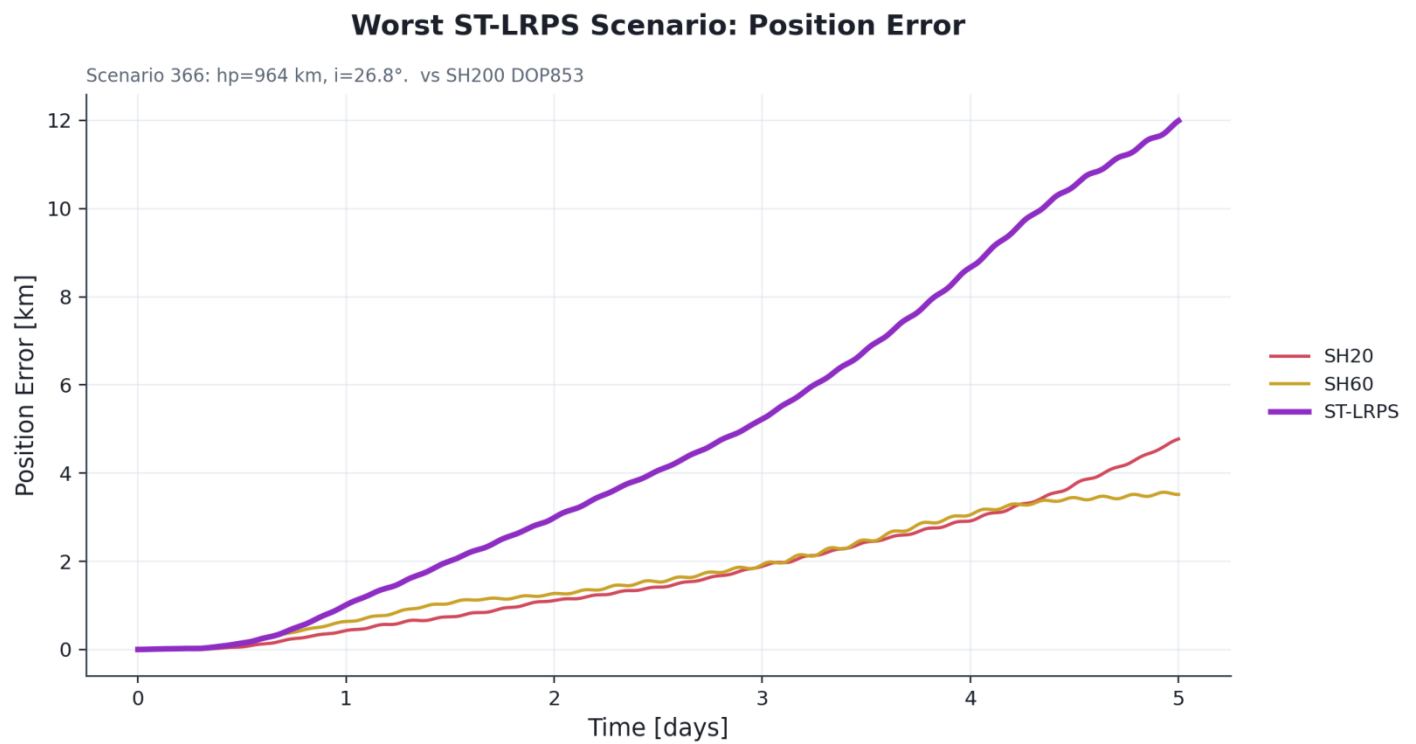
Representative scenario: position error vs time. $N = 100$ scenarios over 5 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



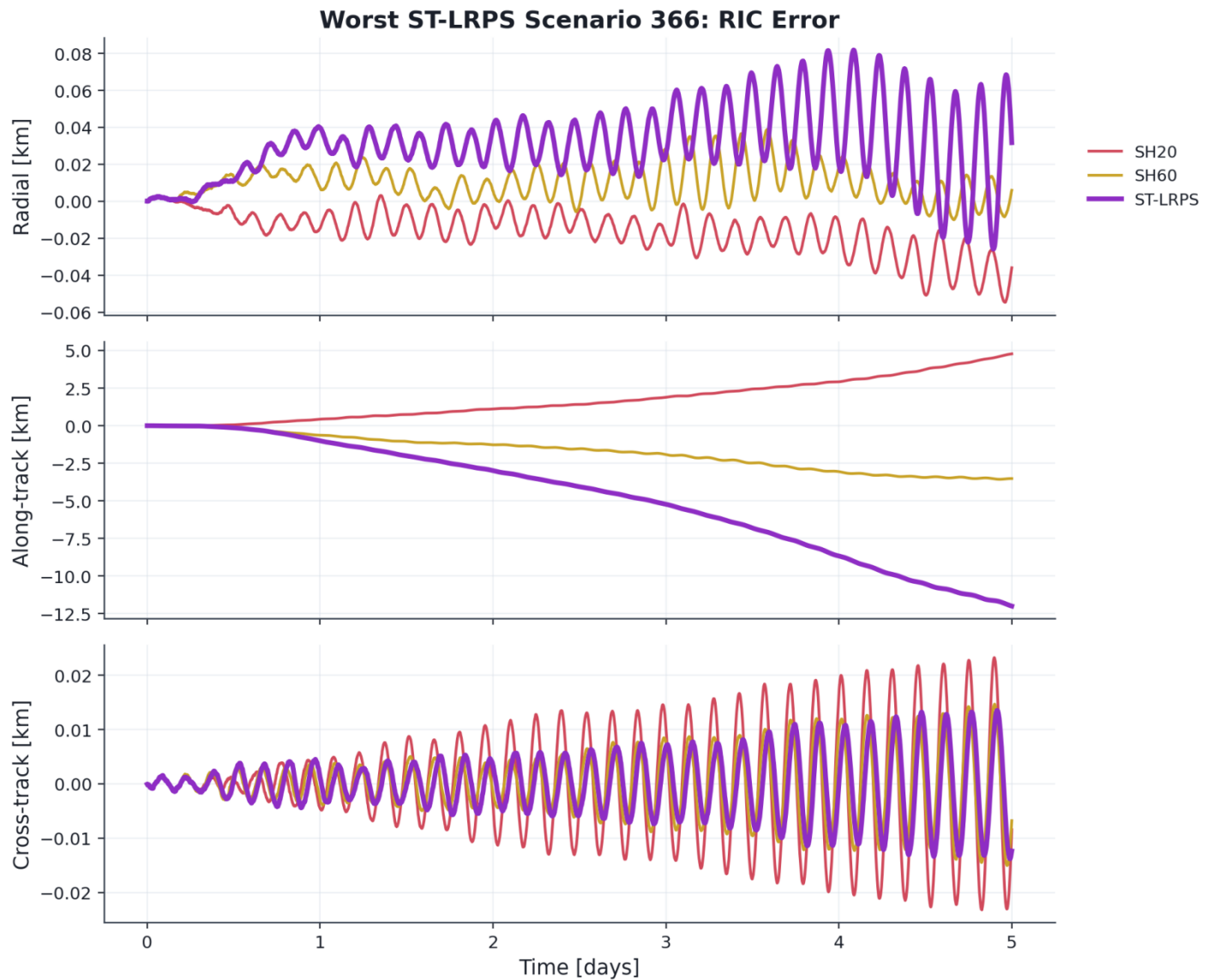
Representative scenario: radial/along-track/cross-track error vs time. $N = 100$ scenarios over 5 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



Worst-case scenario: position error vs time. $N = 100$ scenarios over 5 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Figure



Worst-case scenario: radial/along-track/cross-track error vs time. N = 100 scenarios over 5 day(s); errors are relative to the SH200 DOP853 reference. Axes auto-select display units (km/m/cm); CSV metrics remain in km.

Notes & Caveats

- The truth trajectories are a high-accuracy adaptive numerical reference, not physical ground truth.
- GPU fixed-step SH models include both spherical-harmonic truncation error and integration error.
- ST-LRPS includes surrogate-model error plus integration error.
- Frame mode: `match_dynamics_engine`.